

SLIP – 12

Section I: Data Structures [15 Marks]

Section II: Database Management Systems [15 Marks]

Q 2) Create the following tables (primary keys are underlined).

Sales_order(sorderno, s_order_date, order_amt)

Client(clientno, name, address)

There exists a one-to-many relationship between Client and Sales_order Constraints:
order_amt should be greater than zero.

Q 2(A)

A. Create the above database in PostGreSQL and insert sufficient records. [5 marks]

Answer:

```
CREATE TABLE Client(  
  clientno INT PRIMARY KEY,  
  name VARCHAR(50),      address  
  VARCHAR(100)  
);
```

```
CREATE TABLE Sales_order(  
  sorderno INT PRIMARY  
  KEY,      s_order_date DATE,      order_amt  
  NUMERIC(12,2) CHECK(order_amt > 0),      clientno  
  INT REFERENCES Client(clientno)  
);
```

```
INSERT INTO Client VALUES  
(10, 'Amit', 'Nagpur'),  
(20, 'Neha', 'Pune'), (30, 'Raj', 'Kolhapur');
```

```
INSERT INTO Sales_order VALUES  
(1, '2024-01-10', 5000, 10),  
(2, '2024-03-05', 7000, 20),  
(3, '2024-02-01', 9000, 30),  
(4, '2024-02-10', 1000, 20),  
(5, '2024-02-12', 2000, 20);
```

Q 2(B)

B. Execute the following queries in PostgreSQL (any 3) [6 marks]

i. *Update the client address of all clients from "Nagpur" to "Kolhapur".*

Answer:

```
UPDATE Client
SET address = 'Kolhapur' WHERE
address = 'Nagpur';
```

ii. *Delete all sales order whose client number is 30;*

Answer:

```
DELETE FROM Sales_order WHERE
clientno = 30;
```

iii. *Display all sale records having order date before "23/02/2024".*

Answer:

```
SELECT * FROM Sales_order
WHERE s_order_date < DATE '2024-02-23';
```

iv. *Display sales order having maximum order amount.*

Q 2(C)

C. Write a query to Display client details having maximum sales orders. [4 marks]

Answer:

```
SELECT C.*
FROM Client C
JOIN Sales_order S ON C.clientno = S.clientno
GROUP BY C.clientno
HAVING COUNT(*) >= ALL
(
    SELECT COUNT(*) FROM Sales_order
    GROUP BY clientno
);
```

Q 3) Viva [5 marks]

